Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 8 - Week 8 (Feb 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following statements is/are FALSE regarding the PBFT and PoW consensus mechanisms?

a. PBFT is generally used in permissioned blockchain environments with known validators.
b. PoW is suited for open public networks where participants may be anonymous.
c. PBFT is effective for large, highly decentralized public networks with thousands of nodes.
d. PoW requires significant computational effort to validate new blocks

Answer: (c)

Detailed solution:
That's not correct. Practical Byzantine Fault Tolerance (PBFT) does have quadratic communication overhead ($O(n^2)$), meaning the number of messages grows quickly as more nodes join. As a result, it doesn't scale well to thousands of nodes because every node needs to exchange messages with many others during consensus. In contrast, Proof of Work (PoW) scales better in terms of node count. Nodes don't have to directly communicate with every other node to agree on the state; they simply follow the longest valid chain. While PoW is slower and energy-intensive, it handles large, decentralized networks more effectively than PBFT.

QUESTION 2

Which mechanism ensures deterministic finality in PBFT-based systems?

a. Longest-chain selection
b. Computational difficulty
c. Quorum-based agreement with overlapping honest replicas
d. Randomized leader selection

Answer: (c)

Detailed solution:
PBFT ensures deterministic finality through quorum intersection. Once a block gathers sufficient votes from replicas (at least 2f+1 in a 3f+1 system), conflicting blocks cannot also gather enough votes.

QUESTION 3

In Bitcoin-NG, what is the primary effect of separating leader election from transaction serialization?

a. Eliminates Proof-of-Work
b. Allows higher transaction throughput
c. Makes block intervals deterministic
d. Removes fork possibility

Answer: (b)

Detailed solution:
Bitcoin-NG decouples leader election (via key blocks) from transaction propagation (via microblocks). This enables frequent transaction blocks without increasing mining rate, improving throughput.

QUESTION 4

Which of the following statements is TRUE regarding block propagation in Bitcoin-NG?

a. Micro-blocks are used to carry transactions and are produced frequently between key-blocks.
b. Key-blocks are produced frequently, while micro-blocks are infrequent.
c. Both key-blocks and micro-blocks are produced at the same rate.
d. Micro-blocks determine the leader, while key-blocks only store transactions.

Answer: (a)

Detailed solution:

Bitcoin-NG splits blocks into two types: **Key-blocks** and **Micro-blocks**. Key-blocks are mined using Proof-of-Work and are only used to elect a leader — they don't contain transactions. When a miner wins a Key-block, they become the temporary leader. The leader then creates Micro-blocks, which carry the actual transactions and can be generated quickly without solving new PoW puzzles each time. This way, the network doesn't sit idle between mining events, and transaction processing becomes much faster.

QUESTION 5

What scalability limitation of classical PBFT does CoSi attempt to improve?

a. Energy consumption
b. Quadratic communication overhead
c. Block size restrictions
d. Lack of cryptographic signatures

Answer: (b)

Detailed solution:
Traditional PBFT requires $O(n^2)$ communication between replicas. CoSi introduces tree-based signature aggregation to significantly reduce communication overhead.

QUESTION 6

Which of the following statements is TRUE regarding block (or commitment) propagation in Byzcoin?

a. Byzcoin uses collective signing (CoSi) to allow blocks to be committed quickly once validators reach agreement.
b. Byzcoin relies on a single leader to sign each block, similar to classical PoW mining.
c. In Byzcoin, blocks are only propagated after several minutes to reduce network traffic.
d. Byzcoin avoids using keyblocks entirely and relies only on microblocks for consensus.

Answer: (a)

Detailed solution:
Byzcoin combines ideas from Bitcoin-NG and CoSi. It organizes validators in a communication tree to collect their approvals efficiently. Instead of attaching hundreds of separate signatures, the system aggregates them into one compact Schnorr signature. This single signature proves that a supermajority agreed on the block, which keeps the block small and makes network propagation much faster.

QUESTION 7

In Byzcoin, what is the main benefit of using collective signing for block commitment?

a. Eliminates the need for leader election
b. Reduces signature verification overhead
c. Guarantees probabilistic finality
d. Removes quorum requirements

Answer: (b)

Detailed solution:
Byzcoin uses CoSi to aggregate validator approvals into one collective signature, making block verification efficient and improving scalability.

QUESTION 8

Which trade-off best characterizes BFT-based consensus compared to Proof-of-Work?

a. Higher node scalability but lower throughput
b. No need for identity management
c. Lower communication cost but higher energy consumption
d. Deterministic finality but limited scalability to very large networks

Answer: (d)

Detailed solution:
BFT systems provide deterministic finality and high throughput in smaller networks but scale poorly to thousands of nodes due to communication complexity.

QUESTION 9

In Hyperledger Fabric, why is transaction endorsement performed before ordering?

a. To determine mining difficulty
b. To execute business logic and verify endorsement policy compliance
c. To reduce block interval
d. To eliminate validation

Answer: (b)

Detailed solution:
In Fabric, endorsing peers execute chaincode and verify endorsement policies before transactions are ordered and committed.

QUESTION 10

What happens during the validation phase of a Fabric transaction?

a. Mining competition selects the block
b. The world state is updated without verification
c. Endorsement policies and read-write set consistency are verified before updating world state
d. Proof-of-Work puzzles are solved

Answer: (c)

Detailed solution:
During validation, Fabric rechecks endorsement policies and ensures no conflicts in the read-write set before updating the world state. Only valid transactions modify the ledger state.